

hw9

Ch 18

Problem 1

Let $z = a + b\sqrt{3}i \in \mathbb{Z}[\sqrt{3}i]$ where $a^2 + 3b^2 = 1$

Let $z_2 = a - b\sqrt{3}i$

$zz_2 = (a + b\sqrt{3}i)(a - b\sqrt{3}i) = a^2 + ab\sqrt{3}i - ab\sqrt{3}i + 3b^2 = a^2 + 3b^2 = 1$ from given So then $zz_2 = 1$ so z_2 is the inverse of z

$a^2 + 3b^2 = 1 \implies a = \sqrt{1 - 3b^2}$, since b^2 is always positive, the only integer that keeps the inside of a square root is only positive if $b=0$. Then this means that $a=1$ or $a=-1$

Problem 2

5

Norm(5)= 25 = 5 * 5, Know that $2+i$ has norm of 5 so pick that

$$\frac{5}{2+i} = 5 * \frac{2-i}{(2+i)(2-i)} = 2 - i \text{ so } 5 = (2 - i)(2 + i)$$

1+3i

$$\text{Norm}(1+3i)=1^2+3^2=10$$

So we know that it can be factored into 2 numbers with norm of 2 and 5

Let $a=1+i$ (the only other possibility is $1-i$).

$$\text{So } \frac{1+3i}{1+i} = \frac{(1+3i)(1-i)}{(1+i)(1-i)} = \frac{4+2i}{2} = 2 + i$$

$$\text{So } (1 + 3i) = (1 + i)(2 + i)$$

2

$$\text{Norm}(2)=4=2 * 2$$

So let the first one be $1+i$

$$\frac{2}{1+i} = 2 \frac{1-i}{(1+i)(1-i)} = 1 - i$$

$$\text{So } 2=(1-i)(1+i)$$

6+8i

$6+8i=2(3+4i)$. We already factored 2 above so will skip that for now

Norm($3+4i$)=25= 5 * 5. Pick $2+i$

$$\frac{3+4i}{2+i} = \frac{(3+4i)(2-i)}{(2+i)(2-i)} = \frac{10+5i}{5} = 2 + i$$

$$\text{So } 6+8i=(1-i)(1+i)(2+i)(2+i)$$

Problem 7

Define $\phi : \mathbb{F}(x) \rightarrow D = \{p(x)/q(x) : p(x), q(x) \in \mathbb{F}[x]\}$

Let $a \in \mathbb{F}(x), a = [f(x), g(x)]$ where $f(x), g(x) \in \mathbb{F}[x]$

Def $\phi(a) = \phi[f(x), g(x)] = f(x)/g(x)$. Know that $[f(x), 0] \notin \mathbb{F}(x)$ so $g(x) \neq 0$

Well def:

Let $[f_1(x), g_1(x)] = [f_2(x), g_2(x)]$ then $\phi[f_1(x), g_1(x)] = f_1(x)/g_1(x) = f_2(x)/g_2(x)$ we know because since they are the same equivalence class, that means that $f_1(x)/g_1(x) = f_2(x)/g_2(x)$ which means they are the same frac $= \phi[f_2(x), g_2(x)]$ So this is well def

1-1:

Let $\phi[f_1(x)/g_1(x)] = \phi[f_2(x)/g_2(x)]$

Then $f_1(x)/g_1(x) = f_2(x)/g_2(x) \implies [f_1(x), g_1(x)] = [f_2(x), g_2(x)]$ by how the equivalence class is created which means that $\phi(a) = \phi(b) \implies a = b$ so we are done

Onto:

Let $z \in D$. $z = p(x)/q(x)$ which means that $\phi[p(x), q(x)] = z$ and we know $[p(x), q(x)] \in \mathbb{F}(x)$ so it is onto

Homomorphism:

Let $a = [f_1(x), g_1(x)], b = [f_2(x), g_2(x)]$

$\phi(a + b) = \phi\left[\frac{f_1g_2(x) + f_2g_1(x)}{g_1g_2(x)}\right] = \frac{f_1g_2(x) + f_2g_1(x)}{g_1g_2(x)} = f_1(x)/g_1(x) + f_2(x)/g_2(x) = \phi(a) + \phi(b)$ So

homomorphic

Multiplication is obvious, $\phi(ab) = f_1f_2(x)/g_1g_2(x) = \phi(a)\phi(b)$

For b we can use the same argument

Define $\phi : \mathbb{F}(x_1 \dots x_n) \rightarrow D = \{p(x_1 \dots x_n)/q(x_1 \dots x_n) : p(x_1 \dots x_n), q(x_1 \dots x_n) \in \mathbb{F}(x_1 \dots x_n)\}$

Let $a \in \mathbb{F}(x_1 \dots x_n), a = [f(x_1 \dots x_n), g(x_1 \dots x_n)]$ where $f(x_1 \dots x_n), g(x_1 \dots x_n) \in \mathbb{F}[x_1 \dots x_n]$

Def $\phi(a) = \phi[f(x_1 \dots x_n), g(x_1 \dots x_n)] = f(x_1 \dots x_n)/g(x_1 \dots x_n)$. Know that

$[f(x_1 \dots x_n), 0] \notin \mathbb{F}(x_1 \dots x_n)$ so $g(x_1 \dots x_n) \neq 0$

1-1:

Let $\phi[f_1(x_1 \dots x_n)/g_1(x_1 \dots x_n)] = \phi[f_2(x_1 \dots x_n)/g_2(x_1 \dots x_n)]$

Then $f_1(x_1 \dots x_n)/g_1(x_1 \dots x_n) = f_2(x_1 \dots x_n)/g_2(x_1 \dots x_n)$

$\implies [f_1(x_1 \dots x_n), g_1(x_1 \dots x_n)] = [f_2(x_1 \dots x_n), g_2(x_1 \dots x_n)]$

by how the equivalence class is created which means that $\phi(a) = \phi(b) \implies a = b$ so we are done

Onto:

Let $z \in D$. $z = p(x_1 \dots x_n)/q(x_1 \dots x_n)$ which means that $\phi[p(x_1 \dots x_n), q(x_1 \dots x_n)] = z$ and we know $[p(x_1 \dots x_n), q(x_1 \dots x_n)] \in \mathbb{F}(x_1 \dots x_n)$ so it is onto

Homomorphism:

Let $a = [f_1(x_1 \dots x_n), g_1(x_1 \dots x_n)]$, $b = [f_2(x_1 \dots x_n), g_2(x_1 \dots x_n)]$

$$\phi(a + b) = \phi\left[\frac{f_1 g_2(x_1 \dots x_n) + f_2 g_1(x_1 \dots x_n)}{g_1 g_2(x_1 \dots x_n)}\right]$$

$$= \frac{f_1 g_2(x_1 \dots x_n) + f_2 g_1(x_1 \dots x_n)}{g_1 g_2(x_1 \dots x_n)} = f_1(x_1 \dots x_n)/g_1(x_1 \dots x_n) + f_2(x_1 \dots x_n)/g_2(x_1 \dots x_n) = \phi(a) + \phi(b) \text{ So}$$

homomorphic

Multiplication is obvious

Ch 21

Problem 1

$$\begin{aligned}
 1. & \left(x + \sqrt{\frac{1}{3} + \sqrt{7}}\right) \left(x - \sqrt{\frac{1}{3} + \sqrt{7}}\right) \left(x + \sqrt{\frac{1}{3} - \sqrt{7}}\right) \left(x - \sqrt{\frac{1}{3} - \sqrt{7}}\right) \\
 & \left(x^2 - \frac{1}{3} - \sqrt{7}\right) \left(x^2 - \frac{1}{3} + \sqrt{7}\right) = x^4 - \frac{2}{3}x^2 - \frac{62}{9} \\
 2. & \left(x + \sqrt{3} + \sqrt{5}\right) \left(x - \sqrt{3} + \sqrt{5}\right) \\
 & \left(x + \sqrt{3} + \sqrt{5}\right) \left(x - \sqrt{3} + \sqrt{5}\right) = x^2 - 2x\sqrt{5} - \sqrt{25} = 3 \Rightarrow x^2 - 2x\sqrt{5} - \sqrt{25} = 3 \Rightarrow (x-3)^3 = (2x\sqrt{5} - \sqrt{25})^3 \\
 & \text{this results in } x^6 - 9x^4 - 10x^3 + 27x^2 - 90x - 2 \\
 3. & \left(x + \sqrt{3} + \sqrt{2}i\right) \left(x + \sqrt{3} - \sqrt{2}i\right) \left(x - \sqrt{3} + \sqrt{2}i\right) \left(x - \sqrt{3} - \sqrt{2}i\right) \\
 & = x^4 - 2x^2 + 25 \\
 4. & x = \cos(2\pi/n) + i\sin(2\pi/n) = e^{i(2\pi/n)} \quad x^n = e^{2i\pi} = 1 \quad \text{So } x^n - 1 = 0 \\
 & \text{idk where to go from here, always factorable so DNE} \\
 5. & x = \sqrt[3]{2-i} \Rightarrow x^3 = \sqrt[3]{2-i} \Rightarrow x^3 - \sqrt[3]{2-i} = 0 \Rightarrow x^4 - 2x^2\sqrt[3]{2-i} - \sqrt[3]{4} = 1 \\
 & \Rightarrow x^4 - 1 = 2x^2\sqrt[3]{3} - \sqrt[3]{4} \quad \text{cube both sides \& solve. eventually get} \\
 & x^2 + 3x^8 - 4x^6 + 3x^4 + 12x^2 + 5 = 0
 \end{aligned}$$

Problem 2

1. $\mathbb{Q}(\sqrt{2}, \sqrt{3}) = \{a + b\sqrt{2} + c\sqrt{3} + d\sqrt{6} : a, b, c, d \in \mathbb{Q}\}$ so $\{1, \sqrt{2}, \sqrt{3}, \sqrt{6}\}$ ^{deg 4}
2. Basis of $\mathbb{Q}(\sqrt{2}) = \{1, \sqrt{2}\}$. Basis of $\mathbb{Q}(\sqrt{3}) = \{1, \sqrt{3}\}$ ^{deg 2}
so Basis of $\mathbb{Q}(\sqrt{2}, \sqrt{3}) = \{1, \sqrt{2}, \sqrt{3}, \sqrt{6}\}$ ^{deg 4}
3. $\mathbb{Q}(\sqrt{2}) = \{1, \sqrt{2}\}$ $\mathbb{Q}(i) = \{1, i\}$ so $\{1, i, \sqrt{2}, \sqrt{2}i\}$ ^{deg 4}
4. $\{1, \sqrt{3}, \sqrt{5}, \sqrt{15}, \sqrt{7}, \sqrt{21}, \sqrt{35}, \sqrt{105}\}$ ^{deg 8}
5. $\{1, \sqrt{2}, \sqrt{3}, \sqrt{6}, \sqrt{2}\sqrt{3}, \sqrt{2}\sqrt{6}, \sqrt{3}\sqrt{6}, \sqrt{2}\sqrt{3}\sqrt{6}\}$ ^{deg 6}
6. $\sqrt{8} = 2\sqrt{2}$ so $\mathbb{Q}(\sqrt{8}) \subseteq \mathbb{Q}(\sqrt{2})$ so $\{1\}$ ^{deg 1}
7. $\mathbb{Q}(i, \sqrt{2} + i, \sqrt{3} + i) = \mathbb{Q}(i, \sqrt{2}, \sqrt{3})$ so ^{deg 8}
 $\{1, \sqrt{2}, \sqrt{3}, \sqrt{6}, i\sqrt{2}, i\sqrt{3}, i\sqrt{6}, i\}$
8. $\mathbb{Q}(\sqrt{2} + \sqrt{5}) = \mathbb{Q}(\sqrt{2}, \sqrt{5})$ since over $\mathbb{Q}(\sqrt{25})$, only adding $\sqrt{2}$
 ~~$\mathbb{Q}(\sqrt{2})$~~ so basis: $\{1, \sqrt{2}\}$ ^{deg = 2}
9. We are in $\mathbb{Q}(\sqrt{3} + \sqrt{5}) = \mathbb{Q}(\sqrt{3}, \sqrt{5})$ ~~$\sqrt{6} + \sqrt{10} = \sqrt{2}(\sqrt{3} + \sqrt{5})$~~
which is in our original domain so we are only adding $\mathbb{Q}(\sqrt{3})$
 $\{1, \sqrt{2}\}$ ^{deg = 2}

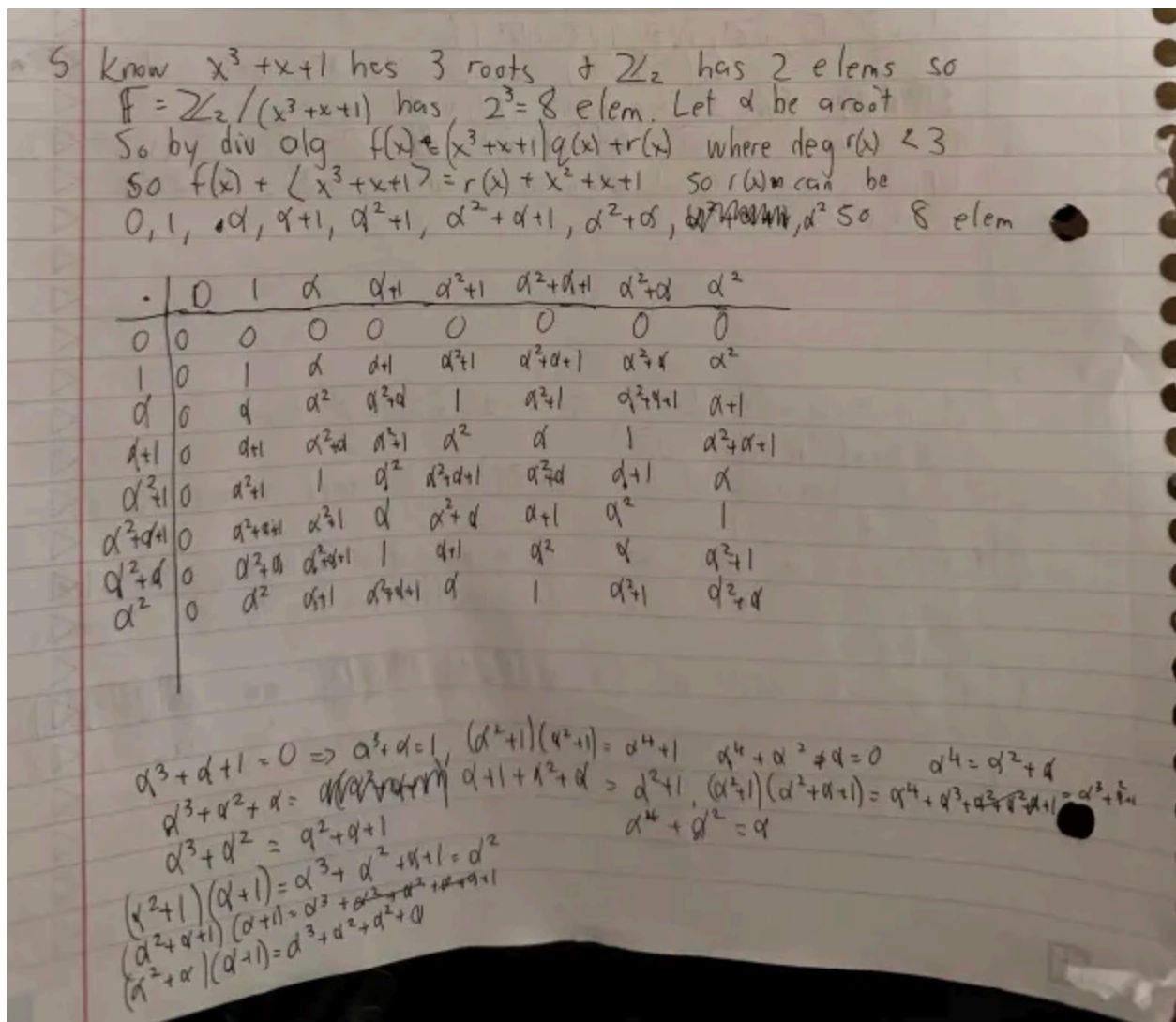
Problem 3

1. $x^4 - 10x^2 + 21 = (x^2 - 7)(x^2 - 3) \Rightarrow x = \pm\sqrt{3}, \pm\sqrt{7}$ so
splitting field: $\mathbb{Q}(\sqrt{3}, \sqrt{7})$
2. $x^4 + 1 = 0$ has roots at $(\pm\sqrt{2} \pm \sqrt{2}i)/2$ so need to add $\sqrt{2} + i$
so $\mathbb{Q}(\sqrt{2}, i)$
3. $x^3 + 2x + 2$ in $\mathbb{Z}_3$ let α be a root then we get $\mathbb{Z}_3(\alpha)$ where
 $f(x) = x^3 + 2x + 2$ has $f(\alpha) = 0$. Since $f(x)$ has deg 3, $3^3 = 27$ elem in
 $\mathbb{Z}_3(\alpha)$
4. $f(x) = x^3 - 3$ roots at $\sqrt[3]{3}, \sqrt[3]{3}\omega, \sqrt[3]{3}\omega^2$ where $\omega = \frac{-1 + \sqrt{3}i}{2}$ so $\mathbb{Q}(\sqrt[3]{3}, i\sqrt[3]{3})$

Problem 4

1. Basis: $\{1, \sqrt{3}, \sqrt{9}, \sqrt{27}, i, i\sqrt{3}, i\sqrt{9}, i\sqrt{27}\}$
8 elem in basis of $\mathbb{Q}(\sqrt[4]{3}, i)$ 1 in $\mathbb{Q}$ so
 $[\mathbb{Q}(\sqrt[4]{3}, i) : \mathbb{Q}] = 8/1 = 8$
2. Obv: $\mathbb{Q}(i)$. Also $\sqrt{9} = \sqrt{3}$ so $\mathbb{Q}(\sqrt{3}) + \mathbb{Q}(\sqrt{3}i)$
3. Obvious ones: $\mathbb{Q}(\sqrt{3}), \mathbb{Q}(i\sqrt{3}), \mathbb{Q}(i, \sqrt{3})$
+ also $\mathbb{Q}(i\sqrt{3} + \sqrt{3}) + \mathbb{Q}(i\sqrt{3} - \sqrt{3})$

Problem 5



Problem 13

Let $\phi : \mathbb{Q}(\sqrt[4]{3}) \rightarrow \mathbb{Q}(\sqrt[4]{3}i)$.

Let $x \in \mathbb{Q}(\sqrt[4]{3}), x = a + b\sqrt[4]{3} + c\sqrt[4]{9} + d\sqrt[4]{27}$

Let $y \in \mathbb{Q}(\sqrt[4]{3}i), y = e + (f\sqrt[4]{3} + g\sqrt[4]{9} + h\sqrt[4]{27})i$ where $a, b, c, d, e, f, g, h \in \mathbb{Q}$

Def $\phi(x) = \phi(a + b\sqrt[4]{3} + c\sqrt[4]{9} + d\sqrt[4]{27}) = a + (b\sqrt[4]{3} + c\sqrt[4]{9} + d\sqrt[4]{27})i$

This is obviously well defined.

Homomorphism:

$$\begin{aligned} \phi(x+y) &= (a+e) + (b+f)\sqrt[4]{3} + (c+g)\sqrt[4]{9} + (d+h)\sqrt[4]{27}i \\ &= a + b\sqrt[4]{3} + c\sqrt[4]{9} + d\sqrt[4]{27} + e + f\sqrt[4]{3} + g\sqrt[4]{9} + h\sqrt[4]{27}i = \phi(x) + \phi(y) \end{aligned}$$

Similarly,

$$\phi(xy) = (a + b\sqrt[4]{3} + c\sqrt[4]{9} + d\sqrt[4]{27})(e + f\sqrt[4]{3} + g\sqrt[4]{9} + h\sqrt[4]{27}i) = \phi(x)\phi(y)$$

Know that $\{1, \sqrt[4]{3}, \sqrt[4]{9}, \sqrt[4]{27}\}$ is a basis for $\mathbb{Q}(\sqrt[4]{3})$ and $\{1, \sqrt[4]{3}i, \sqrt[4]{9}i, \sqrt[4]{27}i\}$ is a basis for $\mathbb{Q}(\sqrt[4]{3}i)$

$$\phi(1) = 1, \phi(\sqrt[4]{3}) = \sqrt[4]{3}i, \phi(\sqrt[4]{9}) = \sqrt[4]{9}i, \phi(\sqrt[4]{27}) = \sqrt[4]{27}i$$

So this means that ϕ maps the basis of $\mathbb{Q}(\sqrt[4]{3})$ to the basis of $\mathbb{Q}(\sqrt[4]{3}i)$, so it must be 1-1 and onto (proof of this done in 590)

Hence they are isomorphic. They are obviously not equal because $\sqrt[4]{3}$ is in $\mathbb{Q}(\sqrt[4]{3})$ but not $\mathbb{Q}(\sqrt[4]{3}i)$

Problem 15

No it is not a field. We know the theorem doesn't apply because $\mathbb{Z}$ is not a field.

Additionally this would add the roots of $x^3 - 2$ to the set. However we still know that the inverse of the nonzero and non 1 integers aren't in the set. For example $5^{-1}=1/5$ is not in this set because it is not in the integers and it is not a multiple of one of the roots of that polynomial, so therefore the inverse of 5 isn't in the set so it isn't a field

Problem 16

Let b be a root of f . Then $f(b) = 0 = b^p + a$ so $-b^p = a$ this means that $f(x) = x^p - b^p = (x - b)^p$ (can do this because p is prime).

This means that there is a singular root to $f(x)$ which means that if $b \in \mathbb{F}$, then $f(x)$ completely factors (or splits). If $b \notin \mathbb{F}$, then we want to show that $f(x)=g(x)h(x)$ is not possible.

So let $f(x)=g(x)h(x)$ where b is not in the field. Then we know that $g(x) = (x - b)^k$ where $k < p$ and $b^k \in \mathbb{F}$ Then by bezout, there exists a m and n st $1=mp+nk \implies b = b^{mp}b^{nk}$ and since $b^p, b^k \in \mathbb{F}, b \in \mathbb{F}$ which is a contradiction.

So if $f(x)$ factors, it must split. If it doesn't factor, it is irreducible